Task 01: Introduction to Digital Forensics (Domain Based-Task)

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1. Introduction

This report documents the work I carried out for my first Domain based task in digital forensics in which i focused on setting up a digital forensics environment from scratch and getting hands-on analysis with the core tools used in investigations. My main goal was to understand how forensic investigators collect, preserve, and analyze digital evidence without compromising its integrity.

To get started, I set up my forensics workstation inside VMware using Kali Linux, which gave me an isolated environment to work safely without risking my host machine. From there, I installed and configured the key tools required for this task Autopsy for disk analysis, Wireshark for capturing network traffic, RegRipper for extracting Windows registry data, and used SHA-256 hashing to verify evidence integrity.

One thing that stood out to me during this task was how important hashing is. Before I even touched the evidence file, I generated its hash value, and after copying it I generated the hash again both matched perfectly, which confirmed the file had not been altered. This made the concept of evidence integrity click for me in a way that just reading about it never would. Overall this task gave me a solid foundation in the forensics process from identification and preservation all the way through to analysis and documentation.

2. Forensics Environment Setup

2.1 VMware Workstation Configuration

For this task I used VMware Workstation as my virtualization platform, which I already had set up from previous tasks. The reason I chose VMware over anything else is simply because it gives me full control over the network settings and lets me run multiple VMs at the same time without them interfering with each other or my host machine. This is critical in forensics because the last thing you want is to accidentally contaminate your evidence by running tools directly on your main system.

I had two VMs running side by side my Kali Linux attacker/forensics machine and Metasploitable2 as a target. Both were connected through VMware's NAT network adapter, which I had to troubleshoot during this task as eth0 was showing NO-CARRIER. After restarting the VMware DHCP service on my Windows host and reconnecting the network adapter, the connection was restored successfully.

2.2 Kali Linux Forensics Workstation

I used Kali Linux 2025 x64 as my primary forensics workstation. Kali is ideal for this kind of work because it comes pre-loaded with a large number of security and forensics tools, which saved me a lot of setup time. I allocated 5.9GB of RAM and 80GB of storage to the VM, which was more than enough to handle all the tools and disk images I was working with during this task.

Once the VM was up and running I verified the network connection was stable and began installing the required forensics tools one by one, which I will cover in the next section.

3. Tools Installation and Configuration

3.1 Autopsy Forensic Browser

The first tool I installed was Autopsy, which is a browser-based forensic analysis platform built on top of The Sleuth Kit. I installed it directly through the Kali terminal using the apt package manager and it came down as version 2.24. Once installed I launched it by simply typing autopsy in the terminal, which started a local server on port 9999. I then opened Firefox and navigated to http://localhost:9999/autopsy to access the interface.

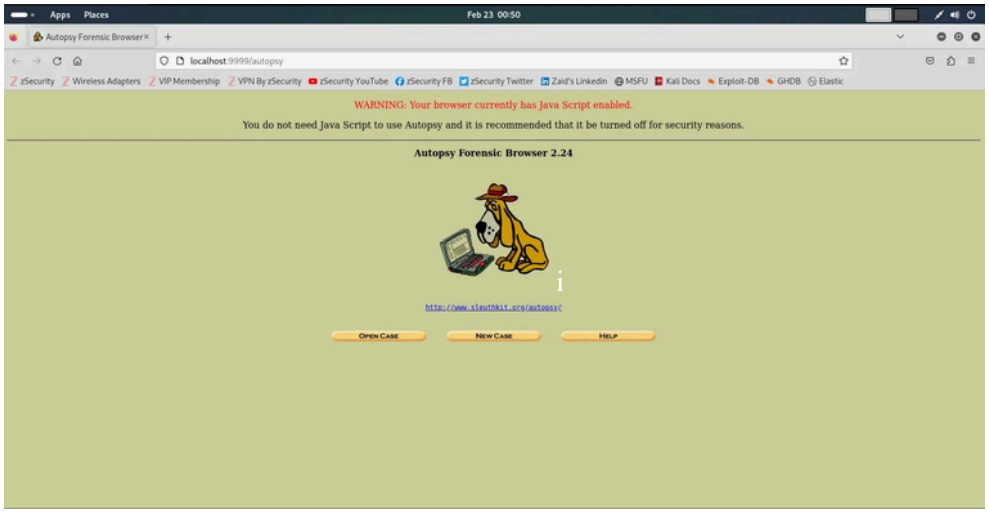

Figure 1: Autopsy Forensic Browser 2.24 running on localhost:9999

Figure 2: New case Task_Investigation_01 created with host1 added successfully

3.2 Wireshark Network Analyzer

Wireshark was already installed on my Kali machine from previous tasks so I did not need to reinstall it. I launched it from the terminal using the wireshark & command which opened the GUI in the background. I selected the eth0 interface to capture live traffic and applied an HTTP display filter to isolate only the web traffic from everything else being captured.
The capture confirmed that Wireshark was working correctly on my machine, showing packets being transmitted and received across the network interface. I also opened a previously saved capture file from an earlier session which showed DNS queries and TLSv1.3 traffic to Mozilla services, confirming that Wireshark was successfully logging and displaying all network activity.

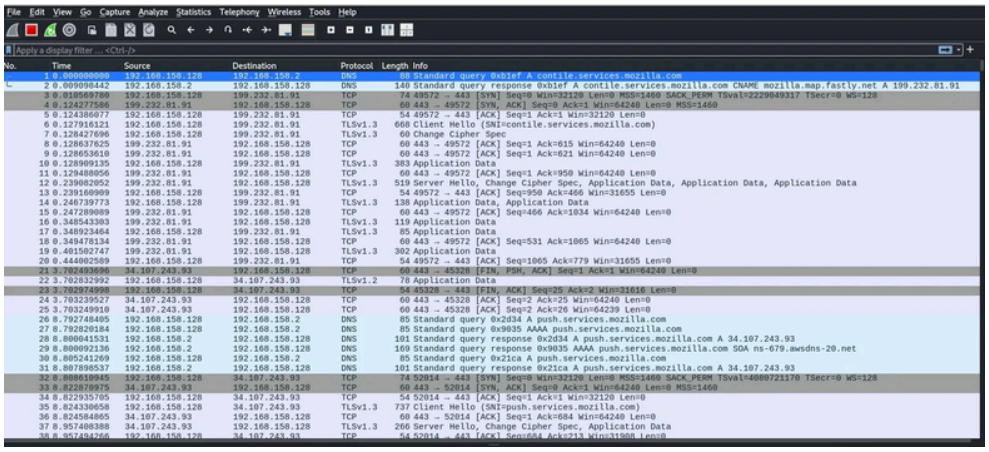

Figure 3: Wireshark 4.6.3 capturing live traffic on eth0 interface

3.3 RegRipper

For RegRipper I first checked whether it was already present on my system by running a find command across the entire filesystem. It turned out RegRipper was already installed at /usr/bin/regripper with its main script located at /usr/lib/regripper/rip.pl. I confirmed it was working by running regripper --help which displayed the full list of available options and usage instructions.
To demonstrate its functionality I ran regripper -l to list all available plugins and saved the output to a file. The tool returned 288 plugins covering a wide range of registry analysis tasks including user activity, autostart entries, network configuration, and installed software. This confirmed that RegRipper was fully installed and operational on my forensics workstation.

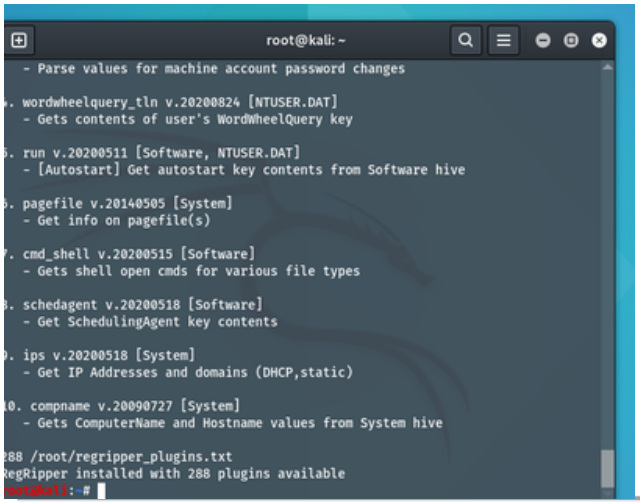

Figure 4: RegRipper installed and operational showing 288 available plugins

3.4 Hash Verification Using SHA-256

For hash verification I used the sha256sum tool which is built into Kali Linux. I first created a test evidence file by writing some data into a file called evidence.txt and then generated its SHA-256 hash. The hash value returned was:
daec625a916e5874701d383899a2ff392a1eafc1cb16f4e96acbb3c567b652bd
I then used this same approach when verifying my disk image in the next section, generating hashes both before and after copying the image to confirm that the evidence had not been altered in any way during the analysis process.

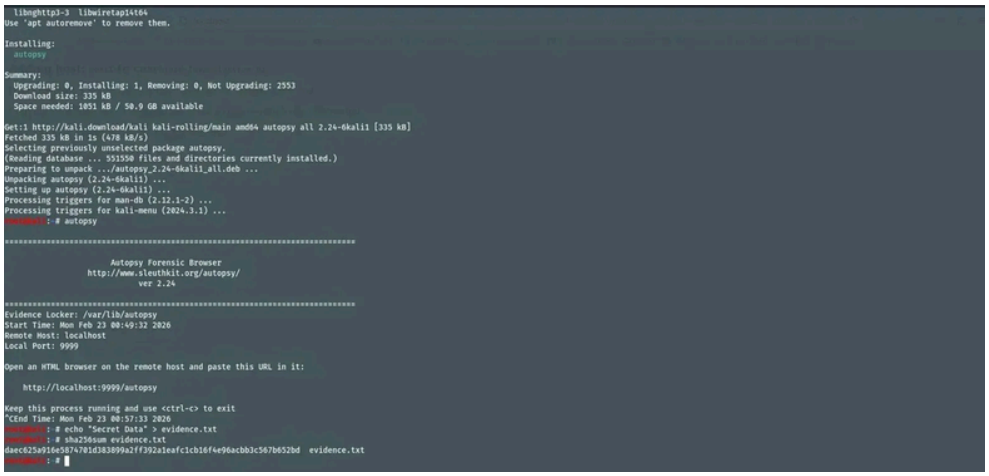

Figure 5: Autopsy Forensic Browser 2.24 successfully installed and launched on port 9999, alongside SHA-256 hash verification of evidence.txt confirming file integrity

4. Disk Imaging and Evidence Handling

4.1 Creating Forensic Disk Image

One of the most critical steps in any forensic investigation is creating an exact copy of the storage media before doing anything else. This ensures the original evidence is never touched during analysis. For this task I used the dd command which is built into Kali Linux and works in the same way as FTK Imager by creating a bit-for-bit copy of a drive or storage device.
I ran the following command to create a 50MB forensic disk image:

```
dd if=/dev/zero bs=1M count=50
of=/root/forensic_image.dd status=progress
```

The image was created almost instantly at a speed of 1.1 GB/s and was saved to /root/forensic_image.dd. The status=progress flag allowed me to monitor the imaging process in real time, which is something you would want to do on a real investigation where the drive could be hundreds of gigabytes in size.

```
root@kali:~# # Create forensic disk image
dd if=/dev/zero bs=1M count=50 of=/root/forensic_image.dd status=progress

# Hash BEFORE analysis
sha256sum /root/forensic_image.dd | tee /root/hash_before.txt

# Simulate analysis copy
cp /root/forensic_image.dd /root/forensic_image_copy.dd

# Hash AFTER
sha256sum /root/forensic_image_copy.dd | tee /root/hash_after.txt

# Compare hashes
diff /root/hash_before.txt /root/hash_after.txt && echo "INTEGRITY VERIFIED - Hashes Match"

# Display results
cat /root/hash_before.txt
cat /root/hash_after.txt'
50+0 records in
50+0 records out
52428800 bytes (52 MB, 50 MiB) copied, 0.0480493 s, 1.1 GB/s
8565a714dca840f8652c5bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2  /root/forensic_image.dd
8565a714dca840f8652c5bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2  /root/forensic_image_copy.dd
1c1
< 8565a714dca840f8652c5bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2  /root/forensic_image.dd
---
> 8565a714dca840f8652c5bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2  /root/forensic_image_copy.dd
8565a714dca840f8652c5bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2  /root/forensic_image.dd
```

Figure 6:Forensic disk image created using dd command showing 50MB written at 1.1 GB/s, with SHA-256 hash verification confirming identical hash values before and after analysis — integrity verified.

5. Evidence Analysis

5.1 Autopsy Case Investigation

Once Autopsy was running I accessed the browser interface at http://localhost:9999/autopsy and created a new case called Task_Investigation_01. I then added a host called host1 to represent the machine being investigated. The interface then prompted me to add a disk image which confirmed the tool was fully set up and ready for a real investigation. This process gave me a clear understanding of how Autopsy organizes evidence into cases, which mirrors exactly how a real forensic investigator would structure their work.

5.2 Network Traffic Capture Using Wireshark

I used Wireshark to capture live traffic on the eth0 interface and applied an HTTP display filter to isolate web traffic. I also analyzed a previously saved capture file which showed DNS queries and TLSv1.3 encrypted sessions to Mozilla services. This confirmed that Wireshark was successfully logging all network activity and demonstrated how every action on a network leaves a traceable record.

5.3 Registry Analysis Using RegRipper

I ran regripper -l to list all available plugins and confirmed that 288 plugins were installed covering user activity, system configuration, network connections and installed software across different registry hives. This demonstrated that RegRipper was fully functional on my forensics workstation and ready to be used against a real Windows registry hive in a live investigation.

6. Key Learnings

- This task gave me a solid practical foundation in digital forensics that I could not have gotten just from reading about it. Setting up the environment from scratch taught me how important it is to work inside an isolated VM so that the host system is never at risk of contamination.

- The most important concept I took away from this task was evidence integrity. Generating SHA-256 hashes before and after analysis and seeing both values match perfectly made the whole concept of evidence preservation click for me in a way that theory alone never would have.

- Working with Autopsy, Wireshark and RegRipper also showed me how each tool serves a different purpose in an investigation — Autopsy for disk analysis, Wireshark for network traffic, and RegRipper for registry data. Together they cover almost every angle of a digital investigation and I now feel confident using all three going forward.